RAG Learning

3.1 Models & Quantities

Mathematics B (MEI) — OCR B — A-Level (H640) — Mechanics — spec refs p31–p35

What this topic covers. The vocabulary of simplifying assumptions in mechanics, the particle model, and the SI base and derived units used throughout the Mechanics component.

1. Why model?

Real situations are too complicated to analyse exactly. A **model** strips away detail judged unimportant, leaving something tractable. The conclusions are then only as good as the assumptions — which is why the specification asks you to state them, and to consider refining them.

2. The language of modelling assumptions

MEI names these words explicitly (spec ref p31).

Assumption	Means	Consequence
Particle	Mass but negligible size	All forces act at one point; rotation is ignored
Light	Negligible mass	The object's weight can be ignored; tension is constant along a light string
Smooth	No friction	No frictional force at the surface or pulley
Rough	Friction present	A frictional force must be included
Inextensible	Cannot stretch	Connected particles share the same magnitude of acceleration

Uniform	Mass evenly distributed	The weight acts at the geometric centre
Thin	Negligible thickness	Treated as a line or a lamina
Rigid	Does not bend or deform	Its shape is fixed under load
Long term	After a long time has elapsed	Transient behaviour has died away

Two of these do specific work in pulley problems. **Light** means the tension is the same throughout the string; **smooth** means the pulley does not change it either, so the tension is equal on both sides. **Inextensible** is what lets you write one acceleration for both particles.

When the particle model breaks down

Modelling a car as a particle is fine when its size is small compared with the distances involved. It is not fine when rotation or the distribution of mass matters — a ladder against a wall, a spinning ball, or a bridge cable whose length is the whole point of the problem. Those need a rigid body.

3. SI base quantities

Quantity	Unit	Symbol
Length	metre	m
Time	second	s
Mass	kilogram	kg

4. Derived quantities

Quantity	Unit	In base units

Velocity	m s^{-1}	m s^{-1}
Acceleration	m s^{-2}	m s^{-2}
Force, weight	newton (N)	kg m s^{-2}
Moment	newton metre (N m)	kg m^2 s^{-2}

A moment is quoted in **N m**, never in joules, even though the base units coincide. The units distinguish a turning effect from an energy.

Conversions

Worked example. Convert 36 km h^{-1} to m s^{-1}.

$$\frac{36\,000\text{ m}}{3600\text{ s}} = 10\text{ m s}^{-1}$$

Going the other way, 5 m s^{-1} is 18 km h^{-1}. A mass of 200 g is 0.2 kg.

5. Mass and weight

Mass is a scalar measured in kilograms and is a property of the object. **Weight** is the *force* of gravity on it, a vector measured in newtons:

$$W = mg$$

A mass of 5 kg has weight $5 \times 9.8 = 49$ N. A mass of 2 kg hanging at rest from a light string produces a tension of 19.6 N.

Taken to the Moon, the object's **mass is unchanged** but its weight falls, because g is smaller there.

The value of g

The acceleration due to gravity is **not** a universal constant — it depends on location. On Earth it is modelled as constant, and the specification states that **unless otherwise specified, examinations use** $g = 9.8$. Using 10 introduces a systematic error of about 2%.

The inverse square law for gravitation is **excluded** from this specification.

6. Scalars and vectors

Scalars	Vectors
Mass, time, distance, speed	Displacement, velocity, acceleration, force, weight

Getting the units wrong is a guaranteed mark loss and is easy to catch: an acceleration quoted in kg, or a force in kilograms, should stop you immediately. Check the units of every answer.

7. Common pitfalls

- Confusing mass with weight, or quoting a weight in kilograms.
- Using $g = 10$ when the specification says 9.8.
- Quoting a moment in joules.
- Applying the particle model where rotation or mass distribution matters.
- Forgetting to convert km h^{-1} to m s^{-1} before using a formula.
- Leaving modelling assumptions unstated when the question asks for them.